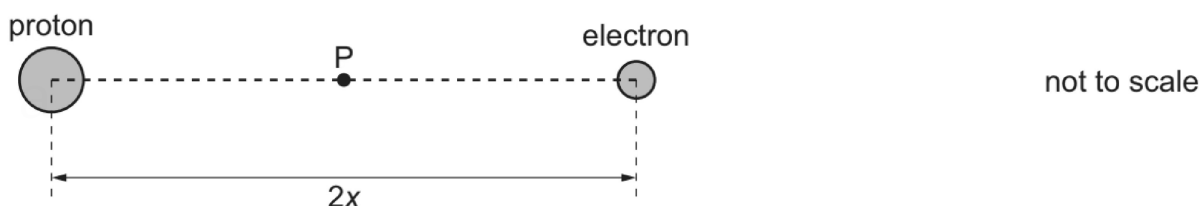


- 4 A hydrogen atom consists of a proton and an electron separated by distance  $2x$ , as shown in Figure 4.1.



**Figure 4.1**

Point P is at the midpoint of the line joining the centres of the proton and the electron.

- (a) (i) For point P, compare the electric field due to the proton with the electric field due to the electron.

.....

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.....

..... [3]

- (ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron.

.....

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.....

.....

..... [3]

- (b) (i) The elementary charge is  $e$  and the permittivity of free space is  $\epsilon_0$ .

State an expression, in terms of  $x$ ,  $e$  and  $\epsilon_0$ , for the magnitude of the electric force  $F$  between the proton and the electron.

$F =$  ..... [2]

- (ii) Determine an expression, in terms of  $F$  and  $x$ , for the electric potential energy  $E_p$  of the proton and the electron.

$$E_p = \dots\dots\dots [1]$$

[Total: 9]